Using World Foundation Models and Digital Twins for Teaching Robot Perception in Higher Education

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***Abstract*—This paper proposes a pipeline that leverages 3D Gaussian Splatting (3DGS) to create a digital twin for generating data augmentations. Designed for an undergraduate robot perception course, the pipeline illustrates the practical application of World Foundation Models (WFMs), such as Cosmos, in generating synthetic data for process variations and hypothetical edge cases. We evaluate the technical feasibility of our proposal by training computer vision models on synthetic image datasets and testing their performance against real images. Additionally, we conducted a study to assess the usability of the approach as a data generation protocol, and to understand how accurately the synthetic images depicting edge cases convey their intended scenarios. Finally, we discuss pedagogical insights and directions to deploy these tools in educational applications.**

***Index Terms*—digital-twin, synthetic data, world foundation model, robot perception, educational innovation.**

I. Introduction

Robotics education involves applying theoretical concepts and methodology within mock environments built using simulators or physical scale models. To maximize learning outcomes, these mock environments should accurately reflect real-world conditions. Although widely adopted open-source simulators offer a practical solution for higher education [1], modeling complex process variations and edge cases remains challenging using traditional toolkits. Establishing an accessible didactic protocol to model those process variations and edge cases can greatly enhance the utility of mock environments, enabling students to gain realistic, practical experience.

Hence, we propose a data augmentation pipeline that leverages photorealistic synthetic data generated by the Cosmos World Foundation Model (WFM) framework. Given a seed scenario, our pipeline generates process variations (domain transfer) and hypothetical events emulating edge cases (predictive inference) as part of a structured didactic protocol for teaching WFM applications in Robotic Perception (RP). This work builds upon our previous efforts, which integrated 3D Gaussian Splatting (3DGS) [2], Unity, and a robotics simulator in a higher-education setting to render photorealistic synthetic dataset variations for training vision models for RP [3], [4].

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To technically validate our approach, we trained a suite of object detection vision models on the augmented synthetic data and evaluated their detection robustness on real robot footage, mimicking a standard Sim-to-Real (Sim2Real) development workflow. Finally, we discuss the impact of structured didactic protocols in robotics education, student learning gains in RP tasks, and the potential of this framework to safely and cost-effectively replicate "5D" scenarios (dangerous, difficult, dirty, dear, and dull) within a classroom environment.

II. Related Works

Recent approaches to graphical simulation heavily leverage Digital Twins powered by 3D Gaussian Splatting (3DGS) [5], [6], a 3D reconstruction methodology capable of photorealistic quality and real-time rendering with native support in modern 3D graphics engines [7]–[9]. For physics simulations, state-of-the-art research increasingly adopts high-end platforms such as NVIDIA IsaacSim and IsaacLab due to their physical fidelity and GPU-accelerated parallelization capabilities [10], [11]. Lately, IsaacSim incorporates plugins to render 3DGS scenes natively, providing a full-stack robotics simulation suite [12].

While IsaacSim is nowadays utilized in industrial contexts, it can face adoption barriers in higher education settings: demanding hardware requirements for student devices, steep learning curve for first-time users, and substantial time required to manually build and tune diverse, realistic virtual scenes. Consequently, educational contexts often favor platforms with broader compatibility across commercial off-the-shelf (COTS) computer hardware and virtual reality (VR) equipment, such as traditional robotics simulators using empirical shading models or game engines like Unity that offer scalable rendering configurations [1], [13], [14].

To provide students with realistic use-case experiences, simulation frameworks must account for environmental process variations and edge cases, which remain notoriously difficult to capture using standard domain randomization in traditional simulators [15]. A prominent solution lies in Physical AI powered by World Foundation Models (WFMs), such as the NVIDIA Cosmos platform [16]. These models perform photorealistic visual domain transfer, predictive inference, and spatial reasoning over multimodal inputs while adhering to physical laws. Although running WFMs locally requires

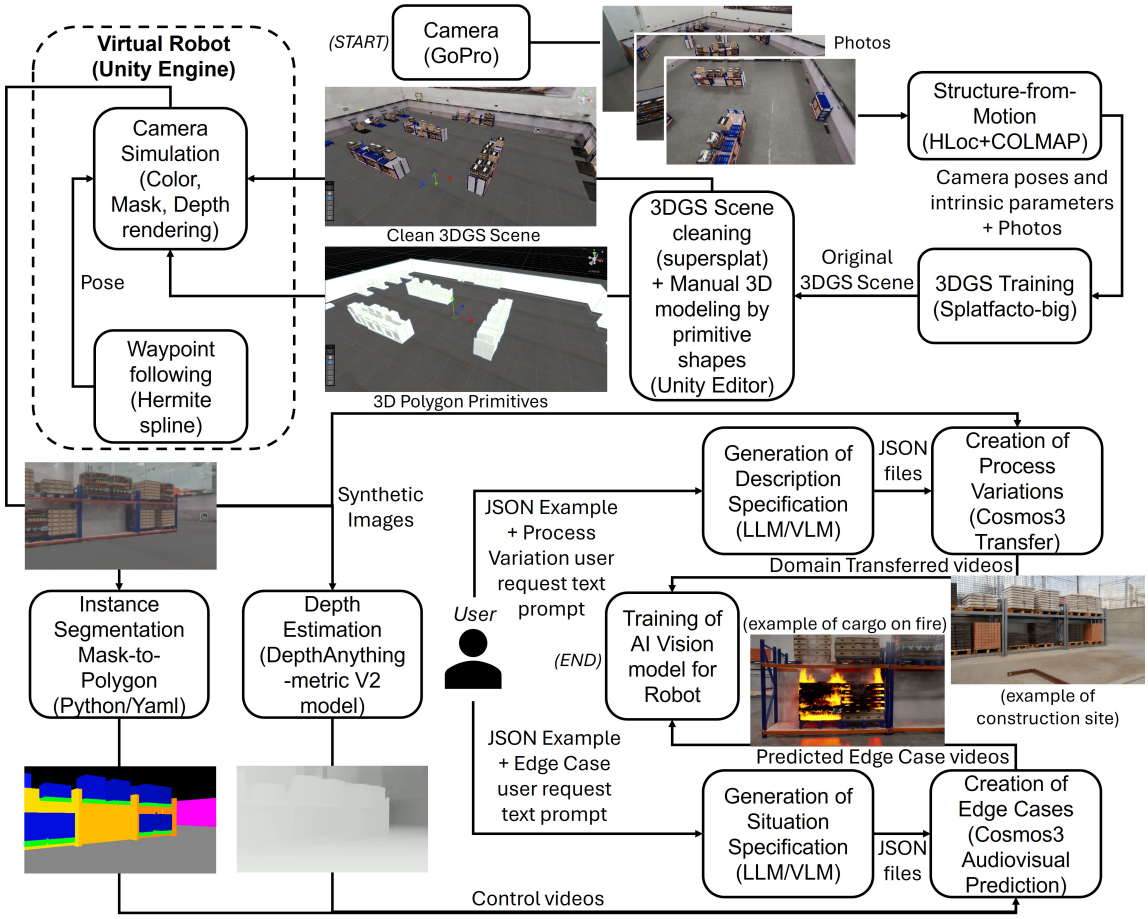


Fig. 1. Overview of the Synthetic Augmented Data Generation Pipeline based on a World Foundation Model

specialized hardware, offline data augmentation via Cosmos can effectively extend traditional robotics simulators and game engines using a limited number of central workstations. Under this hybrid paradigm, an instructor or student can generate high-fidelity process variations and hypothetical edge-case scenarios from a single seed environment, while standard COTS devices handle the dynamic robot physics loop.

To our knowledge, our proposal is the first to integrate WFMs into a higher-education robotics curriculum as an innovation. Technically related efforts in medical robotics include Cosmos-H-Dreams [17], Cosmos-H-Surgical [18]–[21], and the FlashDreams inference server [22], which render photorealistic interactive environments for the learning and training of surgical robot policies. In an industrial context, Saraiva *et al.* [23] leveraged generative models to synthesize training datasets of industrial infrastructure objects (e.g. manometers) via synthetic images with BlenderProc [24] and prompt descriptions, training object detection models for deployment in hazardous environments. In the recent literature, Physical AI via NVIDIA Cosmos is emerging as a foundational component for complex industrial autonomous systems [16], [25].

III. Methodology

Using the framework presented in Fig. 1, we generated a 3DGS digital twin of a physical scale model of a logistics warehouse. The 3DGS scene was reconstructed using `splatfacto-big` within the Nerfstudio framework [26], during 35k epochs, applied to approximately 600 images captured with a GoPro HERO 10 Black at 4k resolution, under a ground-level illuminance range of 450–550 lux. Feature extraction, image matching, and Structure-from-Motion (SfM) camera pose estimation were performed using the Hierarchical Localization (Hloc) toolbox [27], [28] prior to 3DGS optimization. The reconstructed 3DGS scene was manually post-processed and cleaned using SuperSplat [29] before being imported into Unity via the `wuyize25` plugin [7].

To streamline environment setup, primitive geometric shapes (predominantly planes and cuboids) were fitted in the Unity Editor to approximate the coarse scene geometry of the 3DGS scene, including main warehouse assets such as conveyors, racks, pallets, cargo and delivery zones. These primitive shapes primarily served to generate image (instance) masks during the rendering stage using a dedicated shader, and were later converted to polygon masks in YOLO Segmentation format. Additionally, the primitive shapes enabled

the generation of low-fidelity depth maps emulating RGB-D camera outputs using another dedicated shader (grayscale). Finally, pre-defined spatial waypoints were interpolated using a Hermite spline to simulate continuous camera trajectories mimicking autonomous mobile robot navigation.

Then, multi-modal control inputs and textual description prompts were constructed for the Cosmos-3 data augmentation pipeline. All data processing, rendering, and image generation experiments were conducted on a workstation configured with an NVIDIA RTX 6000 Ada Generation GPU (48 GB VRAM), 128 GB RAM, and an Intel Core Ultra 9 processor (24 cores).

A. Multi-modal Control Images

The Cosmos-3 architecture supports diverse spatial control modalities, including world-scenario maps, edge maps, segmentation masks, and depth estimates. In this work, segmentation masks and depth maps were chosen to maintain an abstract visual representation of the scene geometry while preserving the structural identification of primary object classes.

A 25-second seed video sequence (at 24 fps, 640×360 resolution) containing aligned photorealistic color, instance mask, and depth passes was initially rendered in Unity. However, initial iterations with Cosmos-3 revealed that Unity engine-rendered depth maps lacked sufficient detail for optimal model conditioning. Consequently, we adopted DepthAnythingV2-Metric ('base') [30] to infer monocular depth maps directly from the synthetic color frames (Fig. 2). The resulting depth representations had superior structural details, providing a higher-quality visual prior for data generation.

B. Description Prompts

Textual description prompts were synthesized in JSON format using Gemini 3.6 Flash/Think models [31] in conjunction with manual refinement. Prompt structures were adapted from the baseline notebooks provided in the official Cosmos-3 repository [32], specifically utilizing *"run_video_transfer_with_cosmos_framework.ipynb"* for domain transfer and *"run_with_cosmos_framework.ipynb"* for audiovisual predictive inference. The baseline negative prompt configurations were retained without modification.

C. Domain Transfer

For the domain transfer experiments, eight distinct domain variations were established to emulate realistic operational changes: agricultural produce, beverage cargo, e-commerce cardboard packaging, frozen goods, pharmaceutical products, chemical handling, construction materials, and a futuristic Martian settlement environment.

Using the reference JSON template from the original Cosmos/cookbook repository, specialized contextual prompts were issued to the LLM (e.g., *"Now, based on my JSON file, I want a warehouse where construction products are distributed. The pallets should be made of wood, the cargo are cement bags, and bricks of various sizes and colors (grey and red are predominant), the racks should be steel color, the conveyors should be brown color, and the overall scene looks ordered but dusty, while the floor is mostly dirty and the walls are made of concrete with some sections of mesh that shows the construction zone outside of the warehouse."*). The output JSON configurations were passed directly to Cosmos-3 without post-processing. For each of the eight domain variations, four augmented video sequences were generated, yielding 32 unique 25-second videos at 24 fps. Video outputs were cropped by 12 pixels vertically and resized to 854×480 pixels (from an original native inference resolution of 832×480).

D. Predictive Inference

To simulate critical edge-case scenarios from initial static synthetic frames, 4 hypothetical hazard conditions were generated: structural failure of overloaded pallets, cargo falling from high-tier racks, cargo ignition, and dense smoke propagation.

Conditioning prompts were formulated with LLM assistance (e.g., *"Now, based on my JSON file, I want to use an initial image frame and generate a predictive video of 5 seconds that, shows the situation where the cargo starts to catch fire, and the contents start to become dark, all this due to a hypothetical failure when the cargo was checked for safety."*). The generated JSON specifications were supplied directly to the Cosmos-3 predictive model without modification. This yielded four distinct video sequences for each of the four hazard types, producing a total of 16 synthetic edge-case videos (5 seconds each at 24 fps, 832×480 resolution).

IV. Experiments and Results

We evaluate the technical feasibility of our didactic framework for teaching Sim2Real workflows, leveraging Cosmos-3.

A. Process Variations from Domain Transfer

To evaluate the effectiveness of the domain transfer pipeline, we trained two baseline object perception models: RF-DETR [33] (instance segmentation) and YOLO26 [34] (bounding-box object detection). Both models were trained across 7 distinct classes: *cargo*, *pallet*, *rack*, *floor*, *wall*, *delivery*, and *conveyor*. The evaluation encompasses the experimental training scenarios shown in Table I. The following datasets were used: A dataset containing Synthetic images from the 3DGS scene, where the original set included 600 frames (S3), and where the albumentations tools was used to create 4 additional variations for each image (SA); a dataset containing the Augmented images from Cosmos-3 (AC); a simple dataset of 75 frames with Real Images (RI). The resulting performance metrics are summarized in Table II., and a sample detection in Fig. 4.

TABLE I
Training and Testing Scenarios

Scenario	Train Dataset	Train Size	Test Dataset	Test Size
ACvsRI	Augm. Cosmos3 (AC)	19200	Real Images (RI)	75
SAvsRI	Synth. 3DGS Alb. (SA)	3000	Real Images (RI)	75
ACvsS3	Augm. Cosmos3 (AC)	19200	Synth. 3DGS (S3)	600
ACSAvsRI	AC + SA	22200	Real Images (RI)	75

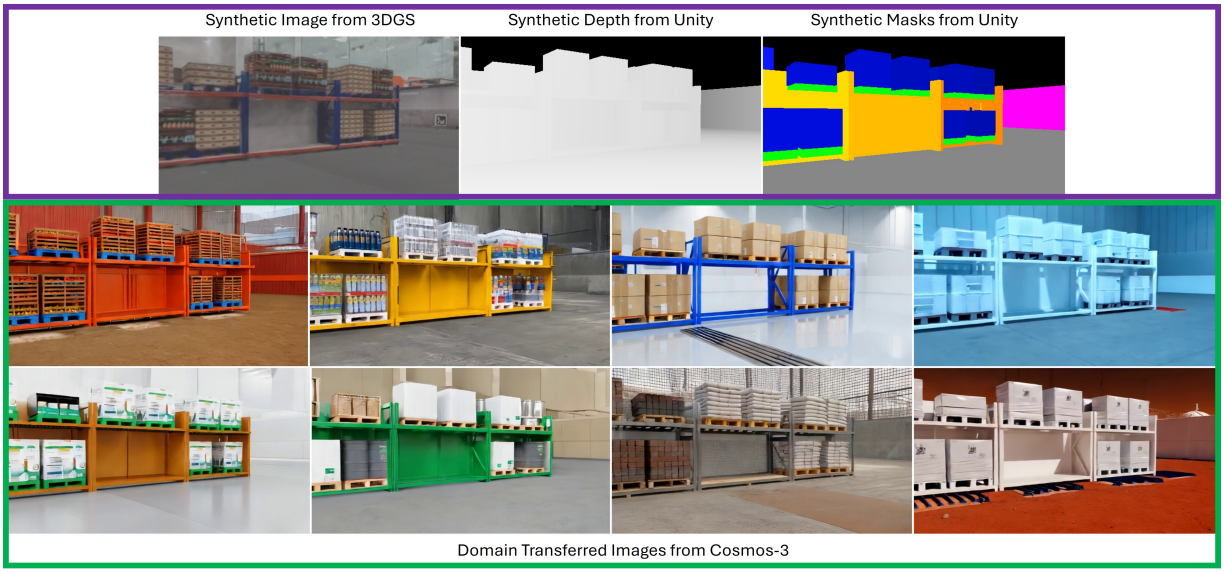


Fig. 2. Representative samples of domain transfer outputs across diverse warehouse operational environments.

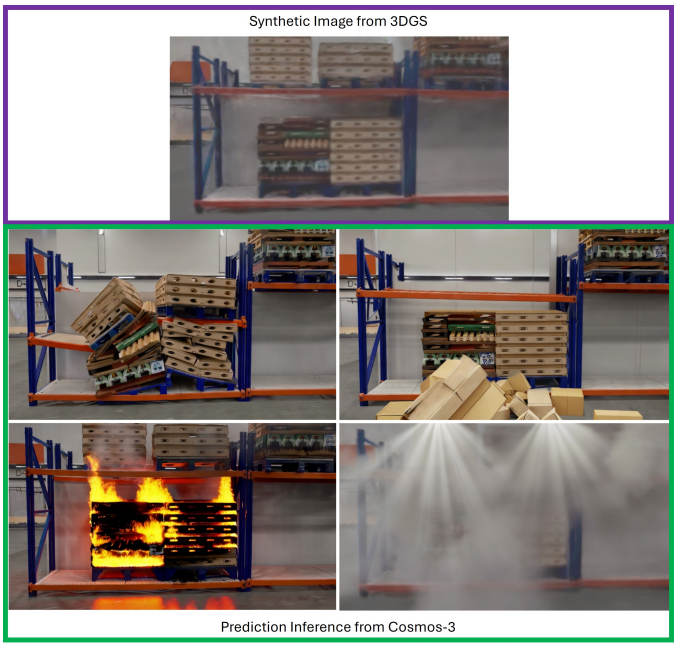


Fig. 3. Predictive inference samples showing edge-case/failure scenarios.

B. Edge Cases from Prediction Inference

A user study was conducted with 10 participants (comprising junior/senior year robotics engineering students, and young industry professionals) who were tasked with evaluating and describing the physical event sequences depicted in the synthetic edge-case videos (either in english or spanish language). Participant observations were categorized across four primary risk scenarios, and qualitative feedback was analyzed using action-verb frequency extraction to identify perceived dynamic behavior and visual artifacts, as summarized in Table III.

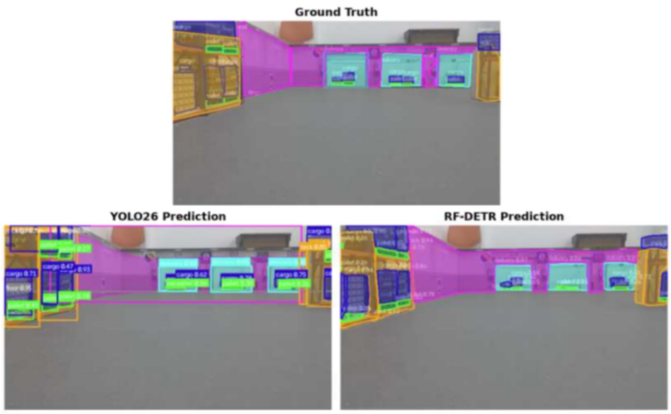


Fig. 4. Object detection sample of the ACvsRI scenario

C. Usability of Data Generation Protocol

A System Usability Scale (SUS) questionnaire [35] was applied to 6 subjects (senior year robotics students), to evaluate our approach as a protocol to generate synthetic data (i.e. lower half of Fig. 1). The subjects were asked to use a video captured by a mobile robot inside our warehouse model and use Cosmos-3 to generate a domain-transferred video (with edge, mask or depth controls) or a predicted edge-case video (from 1 frame). The obtained usability score average was *80.0*.

V. Conclusions

The empirical results validate the technical feasibility of our proposed didactic framework, demonstrating that World Foundation Models (WFMs) such as Cosmos-3 can effectively augment Sim2Real training pipelines for robotics education.

In the real-world evaluation scenarios (*ACvsRI* and *SAvsRI*), models trained on Cosmos-3 domain-augmented data and direct 3DGS synthetic data yielded comparable performance,

TABLE II
DETAILED PERFORMANCE BREAKDOWN ACROSS ACVSRI, SAVSRI, ACVSS3, AND ACSAVSRI SCENARIOS

Model	Class	mIoU	mAP50-95	Precis.	Recall	F1-Score
Scenario: ACvsRI						
RF-DETR	**Overall**	**0.449**	**0.309**	**0.530**	**0.404**	**0.459**
	Cargo	0.706	0.218	0.557	0.321	0.407
	Pallet	0.352	0.056	0.212	0.125	0.157
	Rack	0.292	0.352	0.609	0.528	0.565
	Floor	0.972	0.924	0.929	0.931	0.930
	Wall	0.532	0.213	0.371	0.377	0.374
	Delivery	0.348	0.331	0.803	0.405	0.539
	Conveyor	0.392	0.066	0.233	0.141	0.176
YOLO 26	**Overall**	**0.471**	**0.160**	**0.409**	**0.324**	**0.361**
	Cargo	0.633	0.083	0.422	0.275	0.333
	Pallet	0.357	0.028	0.218	0.121	0.156
	Rack	0.479	0.034	0.225	0.280	0.249
	Floor	0.966	0.894	0.987	0.935	0.960
	Wall	0.569	0.062	0.401	0.305	0.347
	Delivery	0.245	0.013	0.321	0.250	0.281
	Conveyor	0.521	0.008	0.288	0.103	0.151
Scenario: SAvsRI						
RF-DETR	**Overall**	**0.454**	**0.312**	**0.542**	**0.400**	**0.460**
	Cargo	0.679	0.219	0.540	0.307	0.391
	Pallet	0.372	0.052	0.208	0.129	0.159
	Rack	0.376	0.372	0.625	0.520	0.568
	Floor	0.977	0.920	0.929	0.926	0.928
	Wall	0.610	0.270	0.477	0.456	0.466
	Delivery	0.317	0.287	0.775	0.330	0.463
	Conveyor	0.298	0.064	0.243	0.131	0.170
YOLO 26	**Overall**	**0.465**	**0.157**	**0.395**	**0.312**	**0.349**
	Cargo	0.650	0.082	0.386	0.269	0.317
	Pallet	0.359	0.024	0.196	0.121	0.150
	Rack	0.539	0.030	0.176	0.240	0.203
	Floor	0.968	0.890	0.986	0.937	0.961
	Wall	0.496	0.054	0.359	0.274	0.310
	Delivery	0.246	0.007	0.235	0.192	0.211
	Conveyor	0.464	0.014	0.427	0.154	0.226
Scenario: ACvsS3						
RF-DETR	**Overall**	**0.653**	**0.507**	**0.733**	**0.571**	**0.642**
	Cargo	0.832	0.398	0.769	0.489	0.598
	Pallet	0.512	0.157	0.454	0.300	0.362
	Rack	0.849	0.550	0.821	0.604	0.696
	Floor	0.971	0.970	0.986	0.977	0.981
	Wall	0.919	0.759	0.900	0.786	0.839
	Delivery	0.533	0.289	0.537	0.386	0.449
	Conveyor	0.607	0.423	0.667	0.455	0.541
YOLO 26	**Overall**	**0.613**	**0.384**	**0.784**	**0.645**	**0.708**
	Cargo	0.743	0.314	0.816	0.675	0.739
	Pallet	0.432	0.078	0.371	0.382	0.376
	Rack	0.802	0.241	0.744	0.634	0.684
	Floor	0.975	0.970	1.000	0.983	0.992
	Wall	0.845	0.524	0.875	0.697	0.776
	Delivery	0.532	0.234	0.811	0.524	0.637
	Conveyor	0.572	0.330	0.872	0.619	0.724
Scenario: ACSAvsRI						
RF-DETR	**Overall**	**0.517**	**0.370**	**0.615**	**0.449**	**0.519**
	Cargo	0.777	0.348	0.684	0.438	0.534
	Pallet	0.484	0.084	0.317	0.170	0.221
	Rack	0.447	0.425	0.728	0.587	0.650
	Floor	0.960	0.922	0.930	0.927	0.929
	Wall	0.751	0.363	0.527	0.480	0.502
	Delivery	0.379	0.371	0.884	0.404	0.554
	Conveyor	0.340	0.075	0.232	0.137	0.173
YOLO 26	**Overall**	**0.521**	**0.195**	**0.455**	**0.382**	**0.415**
	Cargo	0.727	0.151	0.499	0.340	0.404
	Pallet	0.432	0.032	0.279	0.154	0.198
	Rack	0.532	0.038	0.273	0.360	0.310
	Floor	0.963	0.902	0.996	0.935	0.965
	Wall	0.711	0.220	0.578	0.453	0.508
	Delivery	0.311	0.018	0.373	0.339	0.355
	Conveyor	0.496	0.005	0.190	0.090	0.122

TABLE III
ACTION-VERB FREQUENCY ANALYSIS ACROSS EVALUATED SCENARIOS ($N = 10$)

Scenario	Dominant Action Verbs	Occurrences
Structural failure of overloaded racks	Falling	19
	Crushing	13
	Compressing	10
	Collapsing	8
	Deforming	6
	Passing through	4
Cargo falling from high-tier racks	Falling	18
	Blowing away	10
	Pushing	6
	Throwing	4
	Transforming	3
	Disappearing	2
Cargo ignition and fire	Burning	57
	Spawning	14
	Propagating	7
	Carbonizing	4
Dense smoke propagation	Moving	14
	Tilting upward	12
	Spawning	11
	Covering	5
	Vanishing	5

with no significant advantage observed between the two synthetic generation approaches. Conversely, the cross-synthetic scenario (*ACvsS3*) demonstrated noticeably superior performance metrics compared to both real-world test cases. These findings indicate that training models on purely synthetic data, whether generated via domain transfer through Cosmos-3 or direct digital twin rendering via 3DGS, yields equivalent generalization capabilities when deployed on physical data. Furthermore, the elevated performance in the *ACvsS3* setup suggests that domain-transferred synthetic data may retain underlying structural features and rendering artifacts native to the baseline digital twin, which are inherently shared across synthetic domains but absent in real-world photography. Finally, in the scenario *ACSAvsRI*, we mixed the domain-augmented data with the 3DGS synthetic data for training the models, and we found that the performance was, in general, superior to the *ACvsRI* and *SAvsRI* scenarios. The results of *ACSAvsRI* suggest that when training purely on synthetic data, the overall detection performance improves when we combine both of the synthetic data generation methods.

With respect to the user study, participant observations generally aligned with the intended scenarios depicted in the synthetic edge-case videos. However, several high-frequency verbs potentially indicated recurring instabilities and visual artifacts, specifically with respect to motion-related anomalies (i.e. transforming, disappearing, spawning, and tilting). Handling the observed instabilities may require refined prompt engineering, higher-quality initial frame conditioning, or avoiding edge-case scenarios that exceed the current generation capabilities of Cosmos-3. Finally, the usability score of our proposed synthetic data generation protocol was above the average benchmark of 68 established by Sauro and Lewis [36].

In future work, the synthetic image datasets depicting edge-cases will be utilized to train and evaluate RP models

against real-world risk scenarios. Furthermore, we plan to conduct a formal pedagogical assessment in a larger and live undergraduate classroom setting, employing standardized evaluation questionnaires to measure the efficacy, engagement, and learning outcomes of the proposed didactic protocol. Beyond education, the presented framework may also serve as a practical approach for generating synthetic datasets that represent rare operational conditions and safety-critical events. Such capabilities could support the development and validation of perception systems deployed in automated environments, where collecting real-world examples of these scenarios is often costly, infrequent, or potentially hazardous.

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